ModelIt!

Student Activity Pack

Can a River Eat a Mountain?

The Slow but Powerful Force of Water on Rock

Grade 4 | NGSS 4-ESS2-1 | Cause and Effect

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **Have you ever seen a river or stream? What did the edges look like?**

2. **Can water break apart a rock? Why or why not?**

3. **What do you think the Grand Canyon looked like before the river carved it?**

4. **Draw what you think happens when fast water hits a rocky cliff.**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Water Flow Speed | Rock Hardness | Erosion Rate

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Gentle Stream

Low water speed, hard rock

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Raging River

Maximum water flow speed

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Soft Rock Canyon

Low rock hardness, moderate water

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Wind Speed — Wind also carries tiny particles that sandblast rock surfaces, especially in dry areas

Ice Formation — Water that freezes in rock cracks expands and splits the rock apart piece by piece

Plant Roots — Plant roots grow into rock cracks and slowly push them apart as they grow bigger

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Can something as soft as water actually destroy something as hard as a mountain? Use evidence from your simulation to support your answers.

1 How can something as soft as water destroy something as hard as rock?

2 Why does the Grand Canyon have layers of different colors on its walls?

3 What would happen to mountains if there was no rain or rivers for millions of years?

4 How is erosion both helpful and harmful to humans?

5 How does your model help explain something that takes millions of years?

8 STEM Challenge: Design an Erosion-Proof Riverbank ModelIt!

YOUR ENGINEERING MISSION

Design a riverbank protection system that slows down erosion and keeps soil and rock in place during heavy rains and fast water flow.

THE SCENARIO

A town's riverbank is washing away, and homes near the river are in danger. The town council needs your engineering team to design a solution that slows erosion without harming the river's ecosystem. Use your model evidence to choose the best approach.

Engineering Design Thinking

1. WILL YOU USE PLANTS, ROCKS, WALLS, OR A COMBINATION TO PROTECT THE BANK?
2. HOW WILL YOUR DESIGN HANDLE BOTH NORMAL WATER FLOW AND FLOOD CONDITIONS?
3. WHAT MATERIALS WILL YOU USE AND WHY?
4. HOW WILL YOU TEST WHETHER YOUR DESIGN ACTUALLY REDUCES EROSION?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

Explain your design using scientific evidence from the simulation:

Real Career Connection

Hydrologists and Geomorphologists study how water shapes the land and design solutions to manage erosion, flooding, and water resources. They earn $60,000–$110,000/year.